practice VII.

1. 利用平方乘算法计算 $7^{67} \bmod 97$

将 67 写为二进制 $67 = 64 + 2 + 1$

$67 \Rightarrow 1000011$

① $1 \times 1 \bmod 97 = 1$ $\quad 1 \times 7 \bmod 97 = 7$

② $7^2 \bmod 97 = 49$

③ $49^2 \bmod 97 = 73$

④ $73^2 \bmod 97 = 91$

⑤ $91^2 \bmod 97 = 36$

⑥ $36^2 \bmod 97 = 35$ $\quad 35 \times 7 \bmod 97 = 51$

⑦ $51^2 \bmod 97 = 79$ $\quad 79 \times 7 \bmod 97 = 68$

所以 $7^{67} \bmod 97 = 68$